24JE0609

Project_2(A)

```c
int copy_and_delete(const char *source_path, const char *dest_path) {
    int source_fd, dest_fd;
    ssize_t bytes_read, bytes_written;
    char buffer[BUFFER_SIZE];

    // Start reading the original file
    source_fd = open(source_path, O_RDONLY | O_BINARY);
    if (source_fd < 0) {
        perror("custom_mv: Error opening source file");
        return 1;
    }

    // Create the destination file (or wipe it if it already exists)
    dest_fd = open(dest_path, O_WRONLY | O_CREAT | O_TRUNC | O_BINARY, 0644);
    if (dest_fd < 0) {
        perror("custom_mv: Couldn't open the destination file");
        close(source_fd);
        return 1;
    }

    // Copy the entire file chunk-by-chunk
    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0) {
        bytes_written = write(dest_fd, buffer, bytes_read);
        if (bytes_written != bytes_read) {
            perror("custom_mv: Encountered an error while copying");
            close(source_fd);
            close(dest_fd);
            return 1;
        }
    }

    if (bytes_read < 0) {
        perror("custom_mv: Something went wrong reading the source file");
    }

    // Done copying! Turn off file access
    close(source_fd);
    close(dest_fd);

    // Delete the original file to fully complete the 'move' operation
    if (unlink(source_path) != 0) {
        perror("custom_mv: Copied successfully, but couldn't delete original file");
        return 1;
    }

    return 0;
}
```

This is the code for custom mv

```
nikhil@nikhil-virtual-machine:~/Desktop/hello/OS_Projects_G5/Project_2/custom_commands/test1$ gcc custom_mv.c -o custom_mv
nikhil@nikhil-virtual-machine:~/Desktop/hello/OS_Projects_G5/Project_2/custom_commands/test1$ mkdir subdir
nikhil@nikhil-virtual-machine:~/Desktop/hello/OS_Projects_G5/Project_2/custom_commands/test1$ echo "HELLO WORLD" > test1.txt
nikhil@nikhil-virtual-machine:~/Desktop/hello/OS_Projects_G5/Project_2/custom_commands/test1$ ./custom_mv test1.txt subdir/test1.txt
nikhil@nikhil-virtual-machine:~/Desktop/hello/OS_Projects_G5/Project_2/custom_commands/test1$ ls-l
ls-l: command not found
 127 nikhil@nikhil-virtual-machine:~/Desktop/hello/OS_Projects_G5/Project_2/custom_commands/test1$ ls -l
total 28
-rwxrwxr-x 1 nikhil nikhil 16424 Apr 14 18:40 custom_mv
-rw-rw-r-- 1 nikhil nikhil  2455 Apr 13 02:23 custom_mv.c
drwxrwxr-x 2 nikhil nikhil  4096 Apr 14 18:41 subdir
nikhil@nikhil-virtual-machine:~/Desktop/hello/OS_Projects_G5/Project_2/custom_commands/test1$ tree
.
├── custom_mv
├── custom_mv.c
└── subdir
    └── test1.txt

2 directories, 3 files
```

custom_mv command used to move file to sub directory